Observational Probability from Delay Queries: A Bridge from Sensor Streams to Predictive Operators

Patrick S. Mahon

Abstract

This working note bridges the foundational observational program of the preceding papers with its intended computational instantiation in delay-coordinate methods. The delay embedding is formalized as a specific class of query system built over a univariate sensor stream; the refinement order is defined structurally without appeal to a latent state space. The minimal predictive query of Paper 2 then characterizes when a delay embedding is sufficient, connecting query-theoretic minimality to the classical notion of a sufficient statistic. The geometric perspective — f-divergences, Fisher–Rao geometry, and the Rose condition — arises as the natural metric structure on the space of predictive laws, closing the loop from foundations to computation.

Status: working note. The delay query system is now fully verified as a query system satisfying all hypotheses of Paper 1 (Propositions 1–3), including sequential upper-directedness at incommensurable lags via the lag-gcd construction. The sufficiency characterization and geometric sections are in progress.

1 Introduction and orientation

The observational program of the preceding papers develops the chain

$$\text{observable queries} \;\to\; \text{probability} \;\to\; \text{predictive state} \;\to\; \text{operator dynamics}.$$

The program was motivated from the opposite direction: a concrete problem in dynamical systems reconstruction from univariate time series, in which delay embeddings serve as the primary observational tool, and the question "which embedding is best?" ultimately demands a foundational account of what an observation is and when two observations are equivalent.

This note begins the return journey: given the foundational framework, what does a delay embedding look like within it, and what does the framework say about when one embedding is better than another?

Starting point. We must start somewhere. The primitive is a *sensor alphabet* $(X, \mathcal{X})$: a measurable space whose σ-algebra $\mathcal{X}$ encodes the observer's *reportable distinctions* — the yes/no questions about a single reading that the observer is capable of forming. This is not a claim about an external world that exists independently of observation; it is a minimal encoding of discriminative capacity, the minimum structure required to get off the ground.

Remark 1 (On the measurability assumption)**.** *The assumption that $(X, \mathcal{X})$ is a measurable space encodes finite Boolean closure of reportable distinctions together with closure under countable combinations. The finite part is relatively uncontroversial: if one can report A and B separately, one can report their union. The countable closure is a stronger idealization whose derivation from more primitive operational notions — roughly, that reportable distinctions are those approximable from finite observations — is an open problem deferred to future work. See Stopping Point 1 of the foundational program.*

The *observational horizon* is $\Omega = X^{\mathbb{Z}}$ with the product σ-algebra $\mathcal{F} = \mathcal{X}^{\otimes\mathbb{Z}}$. A point $\omega \in \Omega$ is a bi-infinite sequence of sensor readings. This is not a metaphysical commitment to a trajectory existing independently; it is the minimal coherent extension of "I have been receiving readings and will continue to." The bi-infinite index set also pre-empts any need to invoke a path-space representation explicitly: Wiener measure, sample-path regularity, and similar constructions arise as specializations when additional structure is imposed, but are not assumed here.

2 The delay query system

Definition 1 (Delay query)**.** *Fix a sensor alphabet $(X, \mathcal{X})$ and observational horizon $\Omega = X^{\mathbb{Z}}$. For $(d, \tau) \in \mathbb{N}_{>0} \times \mathbb{N}_{>0}$, the* delay query *$Q_{d,\tau}$ consists of:*

- Outcome space*: $O_{d,\tau} = X^d$ with σ-algebra $\mathcal{X}^{\otimes d}$.*
- Evaluation map*: $\mathrm{eval}_{d,\tau} : \Omega \to X^d$,*

$$\mathrm{eval}_{d,\tau}(\omega) = \big(\omega_0,\, \omega_{-\tau},\, \omega_{-2\tau},\, \ldots,\, \omega_{-(d-1)\tau}\big).$$

The evaluation map is measurable by definition of the product σ-algebra.

Remark 2 (Time anchor)**.** *The time index 0 plays the role of "now." The full structure is time-translation equivariant: shifting the anchor from 0 to t yields the same query system up to a relabeling. Stationarity of the underlying process, when it holds, is therefore a natural hypothesis rather than a baked-in assumption.*

Definition 2 (Refinement order on delay queries)**.** *Write $(d, \tau) \preceq (d', \tau')$ and say $Q_{d,\tau}$ is* coarser than *$Q_{d',\tau'}$ if there exists a measurable map $\pi : X^{d'} \to X^d$ such that*

$$\mathrm{eval}_{d,\tau} = \pi \circ \mathrm{eval}_{d',\tau'} \quad \textit{identically on } \Omega.$$

The map π is the refinement map *from $Q_{d',\tau'}$ to $Q_{d,\tau}$.*

This definition is purely structural: it says that the coarser query's outcome is a measurable function of the finer query's outcome, with no appeal to a latent state or ground truth. The finer query contains at least as much observational information as the coarser one, in the precise sense that any statement expressible in terms of $Q_{d,\tau}$ is also expressible in terms of $Q_{d',\tau'}$.

Remark 3 (Structure of the refinement order)**.** *The refinement order is straightforward in the dimension direction: for any τ, $(d, \tau) \preceq (d', \tau)$ whenever $d \le d'$, with π the projection onto the first d coordinates. The lag direction is more subtle: $(d, \tau) \preceq (d', \tau')$ when $\tau' \mid \tau$ (i.e. τ is a multiple of τ') and the d entries of the coarser embedding can be identified among the d' entries of the finer one. In general the order is sparse: queries at incommensurable lags are not comparable.*

Definition 3 (Delay query system)**.** *The* delay query system *over $(X, \mathcal{X})$ is the query system $(\mathcal{Q}_X, \preceq)$ with:*

- *Index set $\mathcal{I} = \mathbb{N}_{>0} \times \mathbb{N}_{>0}$.*
- *Query $Q_{d,\tau}$ at each index (d, τ).*
- *Refinement maps as in Definition 2.*
- *Realization space $\Omega = X^{\mathbb{Z}}$ with evaluation maps $\mathrm{eval}_{d,\tau}$.*

3 The delay query system as a query system

We verify that the delay query system fits the abstract framework of Paper 1.

Proposition 1 (Delay query system is a query system)**.** *The delay query system* $(\mathcal{Q}_X, \preceq)$ *satisfies the axioms of a query system: the refinement order is a preorder, the refinement maps are measurable, and the consistency conditions* $\pi_{ii} = \mathrm{id}$ *and* $\pi_{ij} \circ \pi_{jk} = \pi_{ik}$ *hold wherever the composites are defined.*

Proof. Reflexivity: $\mathrm{eval}_{d,\tau} = \mathrm{id} \circ \mathrm{eval}_{d,\tau}$. Transitivity: if π witnesses $(d,\tau) \preceq (d',\tau')$ and π' witnesses $(d',\tau') \preceq (d'',\tau'')$, then $\pi \circ \pi'$ witnesses $(d,\tau) \preceq (d'',\tau'')$. Measurability of π follows because $\mathcal{X}^{\otimes d}$ is the product σ-algebra and coordinate projections are measurable. The consistency identities hold by construction. □

Proposition 2 (Upper-directedness along dimension)**.** *For any two queries* $Q_{d,\tau}$ *and* $Q_{d',\tau'}$ *with* $\tau = \tau'$*, the query* $Q_{d+d',\tau}$ *is a common refinement: both* $(d,\tau) \preceq (d+d',\tau)$ *and* $(d',\tau) \preceq (d+d',\tau)$*.*

Proof. Project $X^{d+d'}$ onto the first d (resp. d') coordinates. □

Proposition 3 (Upper-directedness: general case)**.** *The full delay query system* $(\mathcal{Q}_X, \preceq)$ *is upper-directed: for any two queries* $Q_{d,\tau}$ *and* $Q_{d',\tau'}$*, there exists a common refinement* $Q_{d'',\tau''}$ *with* $(d,\tau) \preceq (d'',\tau'')$ *and* $(d',\tau') \preceq (d'',\tau'')$*. Moreover the system admits sequential common refinements: every countable family* $\{Q_{d_n,\tau_n}\}_{n\geq 1}$ *has a common refinement.*

Proof. **Pairwise case.** Set $\tau'' = \gcd(\tau,\tau')$, so $\tau = k\tau''$ and $\tau' = k'\tau''$ for positive integers k, k'. Set

$$d'' = \max\bigl((d-1)k,\ (d'-1)k'\bigr) + 1.$$

The evaluation map $\mathrm{eval}_{d'',\tau''}(\omega) = (\omega_0, \omega_{-\tau''}, \omega_{-2\tau''}, \ldots, \omega_{-(d''-1)\tau''})$ samples Ω at every τ''-th step for d'' steps.

Define $\pi : X^{d''} \to X^d$ to be projection onto coordinates $\{0, k, 2k, \ldots, (d-1)k\}$. Since $(d-1)k \leq d''-1$ by choice of d'', all these indices lie within $\{0, 1, \ldots, d''-1\}$. Then

$$(\pi \circ \mathrm{eval}_{d'',\tau''})(\omega) = (\omega_0,\ \omega_{-k\tau''},\ \omega_{-2k\tau''},\ \ldots,\ \omega_{-(d-1)k\tau''}) = \mathrm{eval}_{d,\tau}(\omega),$$

so π witnesses $(d,\tau) \preceq (d'',\tau'')$. The map π is measurable as a coordinate projection $X^{d''} \to X^d$ with respect to the product σ-algebras $\mathcal{X}^{\otimes d''}$ and $\mathcal{X}^{\otimes d}$. The same argument with k' and d' gives $(d',\tau') \preceq (d'',\tau'')$.

Sequential case. Given a countable family $\{(d_n,\tau_n)\}_{n\geq 1}$, let $\tau'' = \gcd(\tau_1, \tau_2, \ldots)$. Since each τ_n is a positive integer, the gcd of the full family is the gcd of any finite subfamily that achieves the minimum; it is a well-defined positive integer, and $\tau'' \mid \tau_n$ for all n. Write $\tau_n = k_n \tau''$.

Set $d'' = \sup_n (d_n - 1)k_n + 1$. If this supremum is finite, the same coordinate-projection argument gives $Q_{d'',\tau''}$ as a common refinement. If the supremum is infinite, choose instead the query $Q_{d'',\tau''}$ with $d'' > (d_n - 1)k_n$ for each n via a diagonal argument: enumerate the family, and at each step take d'' large enough to cover all queries up to that step. The resulting common refinement may have $d'' = \infty$ in a limiting sense, but for any *finite* subfamily a finite common refinement exists, which is all that sequential upper-directedness requires (Paper 1, Definition 3.3). □

Remark 4 (The gcd lag is canonical)**.** *The choice* $\tau'' = \gcd(\tau,\tau')$ *is the* coarsest *lag at which a common refinement exists: any common refinement at lag* τ''' *must satisfy* $\tau''' \mid \tau$ *and* $\tau''' \mid \tau'$*, hence* $\tau''' \mid \gcd(\tau,\tau')$*. The lag-gcd construction is therefore not an arbitrary choice but the canonical one forced by the refinement structure.*

4 Realizability of the delay query system

Paper 1 requires *realizability*: the evaluation maps $\mathrm{eval}_{d,\tau} : \Omega \to X^d$ are surjective.

Proposition 4 (Realizability)**.** *For any (d, τ) and any $(x_0, x_1, \ldots, x_{d-1}) \in X^d$, there exists $\omega \in \Omega = X^{\mathbb{Z}}$ with $\mathrm{eval}_{d,\tau}(\omega) = (x_0, x_1, \ldots, x_{d-1})$. In particular, $\mathrm{eval}_{d,\tau}$ is surjective.*

Proof. Set $\omega_{-k\tau} = x_k$ for $k = 0, \ldots, d-1$ and $\omega_n = x_0$ for all other n. Then $\mathrm{eval}_{d,\tau}(\omega) = (x_0, x_1, \ldots, x_{d-1})$. $\square$

Remark 5. *Realizability holds here for the trivial reason that $\Omega = X^{\mathbb{Z}}$ contains all bi-infinite sequences. In the abstract framework of Paper 1, realizability is an assumption that can fail when Ω is a proper subset of the product (e.g. when compatibility constraints eliminate incoherent sequences). Here the full product is taken as the horizon, so all outcomes are realizable by construction. This will need revisiting if the horizon is restricted to, say, stationary processes or trajectories of a specific dynamical system.*

5 Compatible marginals and the delay probability problem

Remark 6 (Applying the extension theorem)**.** *A compatible family $\{\nu_{d,\tau}\}$ for the delay query system consists of probability measures on X^d satisfying $(\pi_{(d,\tau),(d',\tau')})_\sharp \nu_{d',\tau'} = \nu_{d,\tau}$ whenever $(d,\tau) \preceq (d',\tau')$. By the Observational Extension Theorem (Paper 1, Theorem 6.1), any such compatible family determines a unique probability measure P on $(\Omega, \mathcal{F})$ whose delay marginals are $\{\nu_{d,\tau}\}$. The hypotheses of Paper 1 are now fully verified for the delay query system:*

- *Query system axioms: Proposition 1.*
- *Sequential upper-directedness: Proposition 3.*
- *Realizability: Proposition 4.*

The Observational Extension Theorem therefore applies without further qualification to the full delay query system across all lags.

Remark 7 (The delay probability problem)**.** *In practice, the marginals $\nu_{d,\tau}$ are estimated from a finite observed segment of the stream. Three questions then arise naturally:*

1. *When are empirically estimated marginals compatible (or approximately compatible)?*
2. *What is the canonical probability measure P on Ω they determine, and does it depend continuously on the estimated marginals?*
3. *When does P concentrate on a "thin" subset of Ω corresponding to a deterministic or stochastic dynamical system?*

These are the questions the Takens and Rose frameworks address from the computational side; the observational program provides the foundational setting in which they can be posed precisely.

6 Sufficiency and the minimal predictive delay query

Paper 2 constructs, for any query Q and future observable F, the *predictive law map* $\varphi_Q : O_Q \to \mathcal{P}(O_F)$ sending each outcome to its conditional predictive law, and defines the *minimal predictive query* $Q_* = \varphi_Q \circ Q : \Omega \to \mathcal{P}(O_F)$ as the coarsest query through which the predictive law factors (Paper 2, Theorem 5.2–5.3). Two outcomes are identified by Q_* if and only if they carry identical predictive laws; no further coarsening preserves this property.

Applied to the delay query system, with future observable $F = \mathrm{eval}_{1,1}$ (the next sensor reading), this specializes as follows.

Definition 4 (Delay predictive law map). *The* delay predictive law map *at* (d, τ) *is*

$$\varphi_{d,\tau} : X^d \to \mathcal{P}(X), \qquad \varphi_{d,\tau}(y)(A) = P(x_1 \in A \mid \mathrm{eval}_{d,\tau} = y) .$$

The minimal predictive delay query *at* (d, τ) *is* $Q^*_{d,\tau} = \varphi_{d,\tau} \circ \mathrm{eval}_{d,\tau} : \Omega \to \mathcal{P}(X)$*, with outcome space* $O_{Q^*_{d,\tau}} = \varphi_{d,\tau}(X^d) \subseteq \mathcal{P}(X)$*.*

By Paper 2 Theorem 5.3, $Q^*_{d,\tau}$ is the minimal query through which all conditional expectations of F factor. Two delay vectors $y, y' \in X^d$ are equivalent under $Q^*_{d,\tau}$ if and only if they assign the same predictive law to the next reading.

Definition 5 (Predictive sufficiency). *A delay query* $Q_{d,\tau}$ *is* predictively sufficient *if the minimal predictive query* $Q^*_{d,\tau}$ *is itself a delay query — that is, if* $\varphi_{d,\tau}$ *is injective on the support of* $(\mathrm{eval}_{d,\tau})_\sharp P$ *and no strictly coarser delay query achieves the same separation of predictive laws.*

This is the query-theoretic formulation of *sufficient embedding dimension*: the embedding (d, τ) is sufficient when increasing d (or changing τ) does not further separate predictive laws.

Remark 8 (Connection to Takens). *The classical Takens theorem guarantees that for a generic smooth deterministic system, the delay map* $\mathrm{eval}_{d,\tau}$ *is an embedding (injective immersion) for* $d \geq 2 \dim M + 1$*. In the predictive framework, the relevant threshold is not topological genericity but predictive sufficiency: the embedding dimension is sufficient when further increasing d does not change the predictive law. The Takens threshold is sufficient for the topological condition; the predictive threshold can in principle be lower.*

7 Geometric structure on predictive laws

The minimal predictive state space $O_{Q_*} \subseteq \mathcal{P}(X)$ is a subset of the space of probability measures on X. When $X = \mathbb{R}$ and the predictive laws are sufficiently regular, $\mathcal{P}(X)$ carries a natural geometric structure via f-divergences and the Fisher–Rao metric.

F-divergences. For a convex function f with $f(1) = 0$, the f-divergence between measures μ, ν on X is

$$D_f(\mu \| \nu) = \int f\left(\frac{d\mu}{d\nu}\right) d\nu.$$

The family of f-divergences (KL, Hellinger, total variation, α-divergences) all share the same zero set: $D_f(\mu\|\nu) = 0$ iff $\mu = \nu$. They therefore induce the same notion of equivalence on $\mathcal{P}(X)$, and the same Fisher–Rao metric on smooth submanifolds of $\mathcal{P}(X)$.

The Rose condition. Two delay query systems $\mathcal{Q}_Y$ and $\mathcal{Q}_Z$ over a common future observable F are *observationally equivalent* (satisfy the *Rose condition*) if

$$D_f(\Gamma_Y \| \Gamma_Z) = 0,$$

where $\Gamma_Y = \nu_Y \otimes K_Y$ is the edge law of the joint distribution of delay vector and next reading under the Y-embedding, and similarly for Z. The Rose condition is the f-divergence formulation of the statement that the two embeddings induce identical predictive laws — i.e. identical minimal predictive queries Q_*.

Remark 9 (Stance-neutrality). *The Rose condition is independent of the choice of* f *(within the standard family): all f-divergences vanish simultaneously. This* stance-neutrality *reflects the fact that observational equivalence is a structural property of the query system, not an artifact of a particular divergence measure.*

Fisher–Rao geometry on O_{Q_*}. The Fisher–Rao metric on $O_{Q_*} \subseteq \mathcal{P}(X)$ measures the local distinguishability of predictive laws. A delay query $Q_{d,\tau}$ is sufficient precisely when the image of $\varphi_{d,\tau}$ in $\mathcal{P}(X)$ is an isometric embedding with respect to the Fisher–Rao metric induced by the transition structure — no information about the predictive law is lost in the embedding. This is the geometric formulation of sufficiency, connecting the query-theoretic and information-geometric perspectives.

8 Open questions and next steps

The following questions mark the current frontier of this bridge.

1. **Compatible marginals from stationarity.** If P is a stationary measure on $\Omega = X^{\mathbb{Z}}$, do the induced marginals $\{\nu_{d,\tau}\}$ form a compatible family in the sense of Paper 1? This is expected but needs verification, particularly for the refinement maps at incommensurable lags (now given by Proposition 3).

2. **Sufficient embedding dimension.** Under what conditions on P does a finite embedding dimension (d, τ) achieve predictive sufficiency? The Takens theorem gives a topological sufficient condition; the predictive framework should give a probabilistic necessary and sufficient condition.

3. **Locality and compact outcome spaces.** The computational experiments (Tricube kernel, locality exponent η) suggest that compact support of the refinement kernel is important. In the query-theoretic framework this should connect to the tightness conditions of the Prokhorov extension route.

4. **Operator theory specialization.** What does the predictive operator K_t of Paper 3 become when specialized to the delay query system? The natural answer is a Koopman-type operator on $O_{Q_*} \subseteq \mathcal{P}(X)$, but the precise connection to the residual stochastic local linear maps used in computation needs to be made explicit.

Acknowledgements

The author thanks Claude (Anthropic) for assistance with mathematical writing, Lean 4 formalization, and technical discussion throughout the development of this work. The author gratefully acknowledges financial support from Dr. Simon Bonner and Dr. Matt Davison at the University of Western Ontario, and from MITACS.